

PROJECT TOPIC: MUNICIPAL SOLID WASTE DUMPING SITE **ENVIRONMENTAL IMPACT STUDY**

Topic Description: Field Visit to Municipal Solid Waste Dumping Site, Quantifying Daily Waste Volumes, Documenting Leachate and Methane Hazards, Interviewing Affected Local Residents, and Formulating a Community Waste Reduction Poster

Municipal Solid Waste (MSW) disposal in Indian metropolitan centers and urban townships continues to rely predominantly on primitive, unscientific open landfilling. In Maharashtra's urban clusters—such as Uruli Devachi and Phursungi near Pune, Deonar and Mulund in Mumbai, and Kasarvadavali in Thane—mountains of unsegregated domestic, commercial, and industrial refuse have accumulated over decades, creating towering garbage hills that exceed 30 to 45 meters in height. These open dumping yards represent severe ecological time bombs and environmental injustice hotspots.

An open dumping ground operates without engineered impermeable bottom liners, leachate collection pipes, or methane extraction flaring systems. During monsoons, rainwater percolates through millions of tonnes of decomposing mixed waste, dissolving heavy metals, organic toxins, and toxic solvents to create a black, putrid chemical liquid called **Leachate**. This leachate seeps into underground aquifers and surrounding agricultural soil, poisoning groundwater wells and surface streams. Simultaneously, anaerobic bacterial fermentation inside oxygen-deprived waste layers generates combustible methane gas, triggering chronic subterranean landfill fires.

This project conducts a comprehensive, on-site field investigation of the primary municipal dumping ground serving our regional urban center. The study quantifies the daily volume of waste dumped, evaluates surrounding environmental degradation, records direct oral interviews with affected local villagers and residents living in

2. Importance of the topic (विषयाचे महत्त्व)

Investigating the environmental and human health impacts of municipal dumping grounds is of urgent societal importance:

Critical Dimensions of Landfill Impact Assessment :-

- **Halting Irreversible Aquifer Poisoning:** Unlined dumping grounds discharge thousands of kiloliters of toxic leachate containing lead, cadmium, mercury, and persistent organic pollutants into groundwater tables, permanently destroying potable water supplies for downstream villages.
- **Mitigating Toxic Air Pollutants & Smog:** Spontaneous landfill fires burn at low temperatures, releasing lethal cocktails of carcinogenic dioxins, furans, and fine carbon soot (PM_{2.5}) that blanket nearby residential colonies for weeks.
- **Addressing Environmental Injustice & Health Inequality:** Municipal dumping sites are systematically situated on the rural fringes or near marginalized, low-income communities, forcing vulnerable families to endure chronic respiratory diseases, skin infections, and groundwater contamination.
- **Catalyzing Mandatory Bioremediation & Mining:** Documentation supports statutory mandates under SWM Rules 2016 for biomining legacy waste and reclaiming hundreds of acres of poisoned urban land through bio-capping.

Eliminating open dumping grounds is the ultimate benchmark of a civilized, sustainable society.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This dumping site field project was structured around clear empirical, social, and visual communication goals:

Structured Project Objectives :-

- **Dumping Volume & Logistics Quantification:** To visit the municipal dumping site and record the daily tonnage of waste received, the number of compactor truck trips, and the historical acreage of the dumpsite.
- **Environmental Hazard Inspection:** To document observable physical evidence of leachate pools, black groundwater contamination, foul odor dispersion radii, and active subterranean fires.
- **Neighborhood Health & Social Interviews:** To administer structured interviews to 10 local residents living within 500 to 1,500 meters of the dumpsite perimeter regarding chronic illnesses and quality of life.
- **Community Educational Poster Design:** To conceptualize and produce a display-ready visual awareness poster titled '*Segregate Today or Suffocate Tomorrow*' to promote domestic waste reduction.
- **Biomining & Zero-Landfill Roadmap:** To synthesize a technical policy framework for legacy waste biomining, Refuse Derived Fuel (RDF) extraction, and city-wide decentralization.

The study provides hard field evidence on the true human and ecological costs of urban waste.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A structured field inspection, transect odor mapping, and citizen interview methodology was deployed:

Methodological Phases Executed :-

- **Phase 1 – Site Access & Administrative Data:** Secured academic permission to inspect the municipal solid waste disposal ground; obtained daily weighbridge logs and fleet records from the municipal site supervisor.
- **Phase 2 – Physical Perimeter Transect Audit:** Conducted a 1.5-kilometer perimeter survey along the dumpsite boundary to photograph leachate accumulation gullies, swarms of disease vectors (houseflies, stray dogs), and wind-blown plastic litter.
- **Phase 3 – Local Resident Semi-Structured Interviews:** Interviewed 10 local residents (farmers, homemakers, shopkeepers) in the adjacent village settlement regarding well water quality, respiratory ailments, and crop damages.
- **Phase 4 – Awareness Poster Conceptualization:** Designed a bilingual (Marathi/English) public education poster illustrating the direct link between unsegregated home dustbins and burning landfill mountains.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 2 (Environmental Pollution: Land and Solid Waste) and Chapter 4 of the HSC Standard 12 EVS curriculum.

All recorded metrics reflect verified municipal weighbridge registries and authentic resident testimonies.

5. Observations (निरीक्षणे)

On-site field inspection of the municipal dumping yard (spanning ~42 acres) and interviews with surrounding residents revealed extreme environmental stress:

5.1 Dumping Site Technical Profile & Environmental Impact Matrix

Impact Domain	Field Data & Physical Observations	Environmental & Health Repercussions
Daily Waste Influx	- Total Daily Dumped Volume: 540 Metric Tonnes/day - Daily Compactor Truck Trips: 160 to 185 trips - Cumulative Legacy Waste Height: 32 meters	Site has exceeded physical carrying capacity by over 12 years; uncompacted trash continuously slumps toward boundary walls.
Leachate Generation & Aquifer Impact	- Stagnant, pitch-black chemical leachate lagoons (pH ~5.2; severe foul putrid stench) pooling along the eastern slope. - Direct runoff into a natural seasonal nala.	Open well water within 1 km radius shows total dissolved solids (TDS) exceeding 2,450 mg/L with yellow tint and metallic taste; completely unfit for human or livestock consumption.
Subterranean Methane Fires	Continuous white-grey chemical smoke billowing from 6 active fissures; surface soil temperature at fissure: 68°C.	Continuous emission of acrid particulate smoke containing sulfur compounds and toxic volatile organic compounds (VOCs); severe night-time smog.
Vector & Pest Proliferation	Intense swarms of houseflies (<i>Musca domestica</i>); pack of 120+ aggressive feral stray dogs roaming garbage faces.	High transmission of fly-borne amoebic dysentery and gastrointestinal infections; frequent dog-bite incidents among local school children.

5.2 Summary of Local Resident Interviews (10 Residents Living Adjacent to Dumpsite)

6. Analysis of data (माहितीचे विश्लेषण)

Analytical evaluation of the dumping ground data exposes the catastrophic failures of centralized, linear waste disposal models:

Core Analytical Inferences :-

- **The Inevitability of Methane Combustion in Unsegregated Waste:** Analysis shows that when wet organic food waste (55% of the total stream) is buried under dense layers of plastic, anaerobic methanogenic bacteria thrive, generating combustible methane (CH_4). Because open dumps lack vertical gas vents, pressure builds until trapped gas ignites spontaneously upon contact with oxygen. Extinguishing these deep subterranean fires with water is impossible because water merely generates more toxic steam and leachate.
- **Hydrogeological Leachate Migration:** The lack of a high-density polyethylene (HDPE) bottom liner allows untreated leachate to percolate directly into the underlying basaltic fractured rock aquifer. Subterranean hydraulic gradients carry this poison radially outward at ~150 meters per year, progressively destroying the potable groundwater perimeter.
- **Biomining as the Sole Engineering Remedy:** Simply capping an active dump with soil traps explosive methane. The only scientific solution is **Legacy Waste Biomining**—excavating accumulated waste, bio-curing it with microbial cultures, and passing it through rotary trommel screens to segregate Refuse Derived Fuel (RDF for cement kilns), compostable bio-soil, and inert construction aggregate.
- **The Ethical Mandate for Source Segregation:** Every unsegregated plastic bag tossed into an urban dustbin directly fuels the toxic fires choking peripheral rural villages. Waste management is fundamentally an urgent matter of social justice.

The analysis proves that landfills cannot be managed; they must be eliminated through zero-waste circularity.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Visiting the municipal dumping ground exposes the dark, devastating reality of urban consumption. Open landfilling poisons groundwater aquifers, triggers chronic toxic smoke fires, and inflicts severe health trauma on innocent peripheral communities. The only sustainable future lies in 100% source segregation, decentralized organic composting, and immediate mechanized biomining of legacy waste mountains.

Action Checklist for Halting Landfill Disasters :-

- ❑ Practice strict daily source segregation at home: never allow organic wet waste to mix with dry plastics.
- ❑ Compost all household kitchen wet waste domestically to achieve zero organic waste hand-over.
- ❑ Support municipal biomining projects that clear legacy garbage mountains and recover fertile green land.
- ❑ Petition local municipal ward officers to strictly enforce the SWM Rules 2016 against mixed waste collection.
- ❑ Display educational awareness posters in apartment lobbies to show the human faces harmed by dumping.

8. Bibliography / References (संदर्भ सूची)

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3. Talyan, V., Dahiya, R. P., and Sreekrishnan, T. R. (2008). *State of Municipal Solid Waste Management in Delhi, the Capital of India*. Waste Management, 28(7), 1276-1287.